

Experiment 1:

First, I opened the Proteus software and created a new project. After that, I selected the required components from the component library such as resistors, signal generator, oscilloscope, and connecting wires.

Then I placed the components on the workspace and built **Circuit A** according to the diagram given in the lab manual. I connected the resistors properly using wires and made sure the circuit connections were correct.

After completing the circuit, I added a **signal generator** as the input source. I set the signal generator to produce a **sinusoidal waveform with a frequency of 100 Hz and peak-to-peak voltage of 5 V**.

Next, I connected the **oscilloscope probes**. One probe was connected across the **input signal** and the second probe was connected across **R1 and R2 one by one** to observe the voltage across each resistor.

Then I started the simulation in Proteus. The oscilloscope displayed the waveform of the input signal and the voltages across the resistors. I observed the waveform carefully and noted the **RMS voltage, peak-to-peak voltage, and waveform shape**.

Finally, I recorded the values from the oscilloscope and used them to draw the waveforms and complete the observation table.

Experiment 2:

First, I opened Proteus and created another circuit for **Circuit B** according to the diagram given in the lab sheet. I selected the required components such as three resistors, signal generator, oscilloscope, and connecting wires from the component library.

After placing the components on the workspace, I connected them carefully to form the required circuit. I checked the wiring to make sure there were no connection errors.

Then I inserted the **signal generator** and adjusted its settings. I set the waveform to **sinusoidal**, the **frequency to 200 Hz**, and the **peak-to-peak voltage to 7 V**.

Next, I connected the oscilloscope probes. One probe was connected across the **input signal**, while the other probe was connected across **R1, R2, and R3 one at a time**. I also checked the voltage across the combinations **R1 and R2** and **R2 and R3**.

After setting everything, I ran the simulation. The oscilloscope showed the input waveform and the output voltages across the resistors. I observed the signals and measured the **RMS voltage and peak-to-peak voltage** from the oscilloscope display.

Finally, I noted the readings and used them to draw the waveforms and fill the required tables in the lab report.